

3주차 미적과제

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$$1. (1) y' = \frac{x+1-x+1}{(x+1)^2} = \frac{2}{(x+1)^2} \rightarrow dy = f(x) dx \rightarrow dy = \frac{2}{(x+1)^2} dx$$

$$(2): dy = f(x) dx = \frac{2}{(x+1)^2} \cdot (0.1) = 0.2$$

$$2. \tanh x = \frac{\sinh x}{\cosh x} \rightarrow \sinh x = 12, \cosh x = 13$$

$$3. (1): \frac{3}{x^3} - 2 \cdot \frac{1}{x^4} \rightarrow f(x) = \frac{3}{4}x^{\frac{1}{4}} - \frac{1}{2}x^{\frac{3}{4}} = \frac{3}{2^{\frac{3}{4}}} \left(\frac{3}{4}x^{\frac{1}{4}} - \frac{1}{2} \right)$$

$$x=0, x=\frac{4}{9}$$

$$(2): g'(x) = 3 - \frac{1}{\sqrt{1-x^2}} \rightarrow 3\sqrt{1-x^2} = 1 \quad 1-x^2 = \frac{1}{9} \quad \frac{8}{9} = x^2$$

$$x=0, x=\pm\frac{\sqrt{2}}{3}$$

$$4. f(x) = 1 - 2\sin x \cos x \quad \sin x \cos x = \frac{1}{2} \quad x = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$f(0) = 0$$

$$2\pi$$

$$f\left(\frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{1}{2}$$

$$0$$

$$f\left(\frac{5\pi}{4}\right) = \frac{5\pi}{4} - \frac{1}{2}$$

$$f(2\pi) = 2\pi$$